



IoT BASED AGRICULTURE MONITORING SYSTEM

Smart Sensing • Real-Time Data • Precision Farming

A Technology Overview: Components, Working, Benefits & Challenges

What is an IoT Agriculture Monitoring System?

An IoT agriculture monitoring system connects field sensors, microcontrollers, and cloud software to continuously track soil, crop, and weather conditions. Farmers view live data on a phone or dashboard and can automate irrigation, alerts, and other field actions without walking every acre by hand.



Monitors soil moisture & irrigation needs



Tracks temperature & humidity



Sends data to the cloud in real time



Goal: Reduce water waste, boost yield, and replace manual field checks with continuous, automated monitoring.

System Architecture — How It Works



Closed-loop system: sensors capture data, the cloud analyzes it, and actuators (pumps, fans) can act automatically — with the farmer always kept in the loop.

Core Components — Sensing Layer

The sensing layer gathers raw environmental and soil data directly from the field.



Soil Moisture Sensor

Measures volumetric water content in soil to guide precise, on-demand irrigation.



Temperature & Humidity Sensor (DHT22)

Tracks air temperature and humidity, key inputs for crop stress and disease risk.



Soil pH Sensor

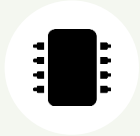
Detects soil acidity/alkalinity to guide fertilizer and lime application.



Rain / Weather Sensor

Detects rainfall and weather shifts to prevent over-irrigation and flooding.

Core Components — Processing, Connectivity & Action



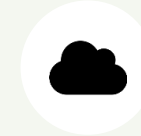
Microcontroller / Board

Arduino, Raspberry Pi, or ESP32 reads sensor data and runs the control logic.



Communication Module

Wi-Fi, GSM/4G, or LoRa modules transmit data from the field to the cloud.



Cloud Platform & Database

Stores historical data, runs analytics, and triggers threshold-based alerts.



Power Supply (Solar + Battery)

Solar panels with battery backup keep remote field units running continuously.



Actuators (Pumps, Valves, Fans)

Execute automated actions such as irrigation or ventilation based on sensor data.



Farmer Dashboard / App

Presents live data, trends, and alerts on mobile or web for quick decisions.

Benefits of IoT in Agriculture



Water Efficiency

Irrigation only when soil actually needs it, cutting water use significantly.



Higher Crop Yield

Data-driven decisions on watering, feeding and treatment improve output.



Lower Labor & Cost

Automated monitoring reduces manual field checks and operating expense.



Real-Time Alerts

Instant notifications let farmers react quickly to stress or weather risk.



Data-Backed Planning

Historical trends support better long-term crop and resource planning.



Sustainable Farming

Optimized input use reduces environmental impact and resource waste.

Advantages (Pros)



Continuous, real-time monitoring of soil and weather conditions



Reduces water and fertilizer waste through precision application



Cuts manual labor needed for routine field inspection



Enables early detection of pest, disease, or stress conditions



Improves crop yield and quality through data-driven decisions



Scalable — works for small farms and large agricultural estates

Disadvantages (Cons)



High initial setup cost for sensors, hardware, and connectivity



Requires stable internet or network coverage in rural areas



Sensors need regular calibration and maintenance for accuracy



Vulnerable to cyberattacks or data privacy breaches



Farmers need basic technical training to operate the system



Battery/power failures can cause data and monitoring gaps

Challenges in Building the System

Key hurdles faced during design, deployment, and long-term operation.



Cost of Hardware

Sensors, gateways, and connectivity modules raise upfront investment, a barrier for small farmers.



Network Connectivity

Remote farmland often lacks reliable Wi-Fi or cellular signal for constant data transmission.



Power Supply

Field units need solar or battery power management to run continuously without grid access.



Data Security

Protecting farm data and preventing unauthorized access to connected devices is critical.



Sensor Durability

Outdoor sensors must withstand rain, heat, dust, and physical damage over long periods.



Data Accuracy & Calibration

Sensors drift over time and need periodic calibration to keep readings reliable.



Conclusion & Future Scope

IoT-based agriculture monitoring turns guesswork into data-driven farming. Despite setup costs and connectivity challenges, it delivers measurable gains in water efficiency, yield, and labor savings — making it a practical step toward sustainable, precision agriculture.



AI-driven predictive analytics for yield & disease forecasting



Drone and satellite imagery integration for field-wide insights



Wider LoRa/5G adoption for low-cost rural connectivity

Thank You